

HR Policy Question-Answering Chatbot using Fine-Tuned Small Language Model

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1. Project Overview

The objective of this project is to develop a domain-specific chatbot capable of answering questions related to Human Resources (HR) policies. Instead of relying solely on a general-purpose Large Language Model (LLM), the project focuses on creating a customized knowledge base from HR policy documents and fine-tuning a Small Language Model (SLM) to provide accurate and context-aware responses.

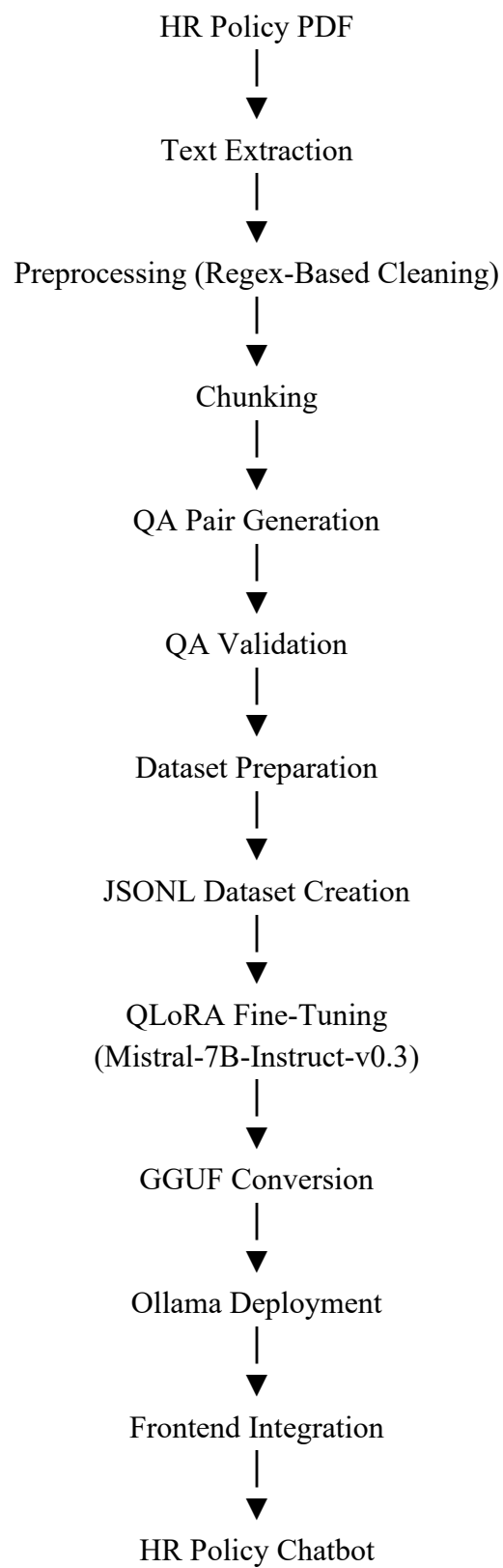
The solution leverages document processing, dataset generation, parameter-efficient fine-tuning (LoRA/QLoRA), and conversational AI techniques to build an intelligent HR assistant.

2. Project Objectives

The primary objectives of this project are:

- Extract and preprocess information from HR policy documents.
- Generate high-quality question-answer datasets from the extracted content.
- Fine-tune a Small Language Model using LoRA-based techniques.
- Evaluate the model's performance on domain-specific HR queries.
- Develop a user-friendly chatbot interface for interaction.

3. Solution Architecture



4. Methodology

Phase 1: HR Policy Data Collection

An HR policy PDF document was selected as the primary source of knowledge. The document contains organizational policies, guidelines, employee benefits, leave rules, and other HR-related information.

Output:

HR Policy PDF

Phase 2: Text Extraction and Preprocessing

The PDF content was extracted using Python-based document processing libraries. The extracted text underwent preprocessing to ensure quality and consistency.

Preprocessing Activities

- Removal of unnecessary spaces and formatting issues.
- Cleaning special characters and unwanted symbols.
- Normalization of text.
- Preservation of meaningful policy content.

Output:

- Clean textual representation of HR policy information.

Phase 3: Document Chunking

The cleaned document was divided into smaller chunks to ensure that the language model receives focused and contextually relevant information during question-answer generation.

Benefits of Chunking

- Improved context management.
- Better QA generation quality.
- Reduced information loss.
- Higher coverage of policy topics.

Output:

- Structured text chunks.

Phase 4: Question-Answer Pair Generation

The chunked HR content was provided to the **Qwen2.5-3B-Instruct** model.

The model was prompted to automatically generate question-answer pairs based on each chunk.

Example

Context: Employees are entitled to 15 casual leave days per calendar year.

Generated Pair:

```
{  
  "question": "How many casual leave days is an employee entitled to per year?",  
  "answer": "Employees are entitled to 15 casual leave days per calendar year."  
}
```

The generated pairs were stored in a structured JSONL format.

Output:

- Initial QA Dataset (.jsonl)

Phase 5: QA Validation

Generated question-answer pairs were reviewed and validated to ensure:

- Relevance to source content.
- Factual correctness.
- Completeness of answers.
- Removal of duplicates.
- Elimination of hallucinated information.

Validation Criteria

- Question correctness
- Answer accuracy
- Context alignment
- Language quality

Output:

- Validated QA Dataset

Phase 6: Model Fine-Tuning

The validated QA dataset was used directly to fine-tune Mistral-7B-Instruct-v0.3.

To reduce computational requirements and memory usage, parameter-efficient fine-tuning was performed using:

- LoRA (Low Rank Adaptation)
- QLoRA (Quantized LoRA)

Reasons for Selecting QLoRA

- Reduced GPU memory consumption.
- Faster training.
- Lower computational cost.
- Comparable performance to full fine-tuning.

Fine-Tuning Process

1. Load base Mistral model.
2. Apply quantization.
3. Configure LoRA adapters.
4. Train on validated HR QA dataset.
5. Save adapter weights.
6. Merge model for inference.

Output:

- Fine-tuned HR Chatbot Model

Phase 7: GGUF Conversion and Local Model Deployment

After fine-tuning, the trained model was converted into the **GGUF (GPT-Generated Unified Format)** to enable efficient local inference.

Purpose of GGUF Conversion

- Reduced model storage requirements.
- Faster inference performance.
- Compatibility with local inference engines.
- Simplified deployment without requiring cloud services.

Output:

- GGUF-compatible HR chatbot model

Phase 8: Local Runtime Integration Using Ollama

The GGUF model was deployed using **Ollama**, which served as the local inference and model runtime layer.

Role of Ollama

- Loads the GGUF model locally.
- Handles model inference requests.
- Provides an API endpoint for frontend communication.
- Eliminates dependency on external LLM APIs.

Benefits

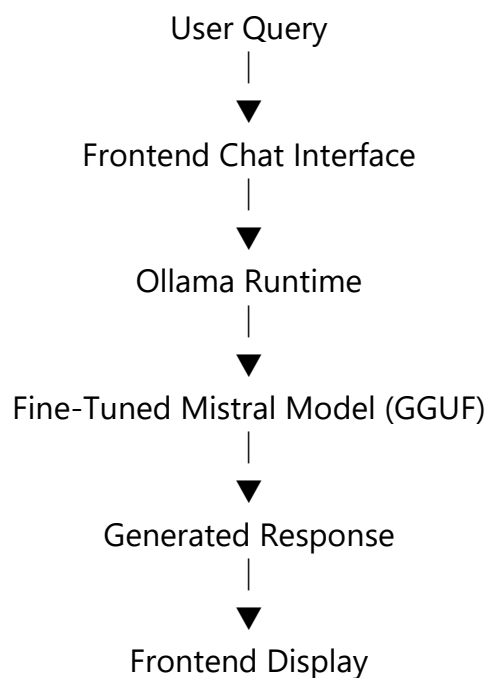
- Fully local execution.
- Improved data privacy.
- Reduced operational costs.
- Faster experimentation and testing.

Output:

- Locally deployable HR chatbot backend

Phase 9: Frontend Integration and Chatbot Development

The frontend application was connected to the locally hosted Ollama runtime to enable conversational interaction with the fine-tuned model.



User Interaction Flow

5. Model Evaluation

After fine-tuning, the model was evaluated using a validation dataset containing HR-related question-answer pairs. **BERTScore** was used to measure the semantic similarity between the model-generated answers and the reference answers. The evaluation focused on the semantic quality and relevance of the generated responses rather than exact word matching.

Evaluation Method:

- BERTScore
- Comparison between generated answers and reference answers
- Semantic similarity measurement
- Evaluation on HR-domain questions

Evaluation Parameters:

- Answer Accuracy
- Semantic Similarity
- Domain Relevance
- Context Understanding
- Response Completeness

Output:

- BERTScore-based performance evaluation of the fine-tuned HR chatbot

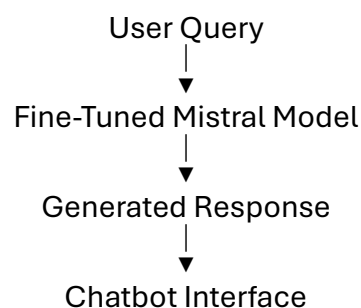
6. Chatbot Development

A chatbot user interface was developed to provide interactive access to the fine-tuned model.

Features

- Natural language interaction.
- HR policy question answering.
- Fast response generation.
- User-friendly interface.

Workflow



7. Technologies Used

Component	Technology
Programming Language	Python
Data Source	HR Policy PDF
Text Extraction	PyMuPDF
Text Preprocessing	Regular Expressions (Regex)
QA Generation	Qwen2.5-3B-Instruct
Dataset Format	JSONL
Fine-Tuning Method	LoRA / QLoRA
Base Model	Mistral-7B-Instruct-v0.3
Model Evaluation	BERTScore
Model Format	GGUF
Local Inference Runtime	Ollama
Frontend	Next.js
Deployment	Local Deployment

8. Observations

- The fine-tuned Mistral-7B-Instruct-v0.3 model was able to generate responses specific to the HR policy domain.
- The model showed better alignment with the terminology and information present in the HR policy dataset after fine-tuning.
- The model was able to answer common HR-related queries such as questions about leave policies, work-from-home policies, holidays, and employee-related policies.
- The responses were generally more focused on the provided HR policy information compared to a general-purpose model.
- Model performance was quantitatively evaluated using BERTScore, as described in the Model Evaluation section.
- The model achieved a BERTScore Precision of 0.9350, indicating that the generated responses contained information that was highly relevant to the corresponding reference answers.
- A Recall score of 0.9178 indicates that the model was able to capture a substantial portion of the information contained in the reference answers.
- The model achieved a BERTScore F1 of 0.9260 (92.60%), showing a high degree of semantic similarity between the generated responses and the reference answers.
- The results indicate that the fine-tuned model was able to generate responses that were generally consistent with the expected HR policy information.
- The slightly lower Recall compared to Precision suggests that, in some cases, the model may provide a more concise response while not capturing every detail present in the reference answer.

9. Challenges and Mitigation

- **Limited Domain Scope:** The chatbot is trained specifically on the available HR policy dataset. Therefore, its ability to answer questions outside the covered HR policies or domain is limited.
- **Synthetic Data Expansion:** Attempts were made to expand the question-answer dataset using synthetically generated questions. However, many generated questions were duplicated or highly similar to existing questions. Consequently, the number of unique and useful training samples did not increase significantly. Moreover, increasing the synthetic data did not result in a noticeable improvement in model accuracy. Therefore, synthetic data expansion was not considered beneficial for the final training dataset.
- **Dependence on Training Data Quality:** The performance of the fine-tuned model depends strongly on the quality, relevance, and coverage of the question-answer dataset. Missing or insufficiently represented policy information may affect the model's ability to provide complete responses.
- **Response Variability:** As a generative language model, the chatbot may occasionally produce responses with different wording or levels of detail for similar questions. Therefore, semantic evaluation was used rather than relying only on exact text matching.
- **Limited Evaluation Dataset:** The model was evaluated using 72 HR policy-related questions. Although the results provide an indication of model performance, a larger and more diverse evaluation dataset could provide a more comprehensive assessment.
- **Computational Resource Constraints:** Fine-tuning and evaluation of the language model require considerable computational resources. Memory limitations can restrict batch size, sequence length, and other training parameters.
- **Local Deployment:** The current implementation is deployed locally using Ollama. It has not yet been tested under large numbers of simultaneous users or in a production-scale environment.
- **Policy Coverage:** The chatbot's responses are limited to the information represented in the source HR policy document and the prepared training dataset. New or updated HR policies would require additional data preparation and model updating to maintain current responses.

Expected Outcomes

- Domain-specific HR chatbot.
- Improved response relevance for HR-related queries.
- Efficient deployment through parameter-efficient fine-tuning.
- Reduced dependence on external retrieval systems.
- Demonstration of end-to-end LLM fine-tuning workflow.

10. Future Enhancements

- Incorporate multiple HR policy documents.
- Implement Retrieval-Augmented Generation (RAG).
- Add multilingual capabilities.
- Introduce feedback-based continuous learning.
- Integrate with Microsoft Teams or web applications.
- Monitor model performance using automated evaluation metrics.

11. Conclusion

This project demonstrates an end-to-end workflow for building a domain-specific conversational AI system. Starting from HR policy PDFs, the solution extracts and preprocesses textual information, generates and validates question-answer pairs, prepares the validated QA dataset for instruction fine-tuning, and fine-tunes a Mistral-7B-Instruct-v0.3 model using LoRA/QLoRA techniques. The resulting chatbot provides HR policy assistance through a specialized and resource-efficient language model, showcasing practical applications of LLM fine-tuning and conversational AI in enterprise knowledge management.